



STUDENT INNOVATION CHALLENGE · TSHWANE UNIVERSITY OF TECHNOLOGY

Corridor Hills.

Digital Maintenance & Residence Operations Platform

A residence-aware digital platform for reporting, managing and resolving maintenance issues.

[REPORT AN ISSUE →](#)

“Maintenance is not only about **fixing** what is broken.”

It is about knowing what is broken, where it is, who should fix it, how long it has remained unresolved, and whether it was actually fixed.

KNOWING

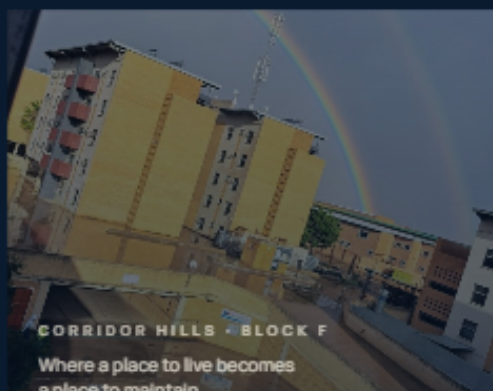
what is broken

LOCATING

where it lives

ASSIGNING

who should fix it



A STUDENT INNOVATION PROPOSAL

Designed for one specific residence, by acknowledging what that residence is actually like.

Tshwane University of Technology

What this proposal is, and why it exists.

Maintenance problems are an unavoidable part of residence life. A broken light, blocked sink, faulty stove, damaged bed or leaking pipe can significantly affect a student's living environment. The challenge is rarely only the physical repair.

THE BIGGER QUESTION

How does a residence efficiently know what is broken, where it is, who should fix it, and whether it was resolved?

THE PROPOSAL

01

Reporting, simplified.

Verified residents report a maintenance issue from any device, in seconds.

02

Work, clarified.

The right technician receives the right job with the right context.

03

Operations, visible.

Leadership sees what is happening across the residence.

WHAT THIS IS NOT

Not an idea board. Not an academic prototype.

A residence-aware operations platform, structured around the actual physical environment.

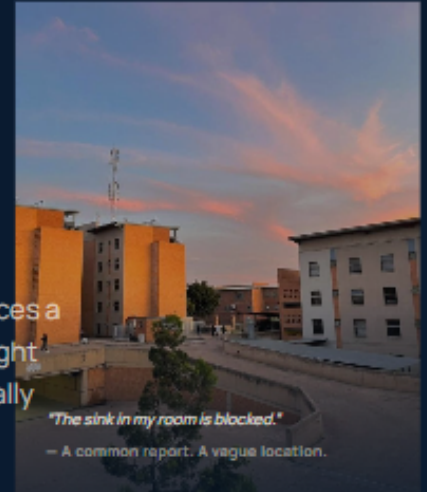
THE OBJECTIVE

Make reporting easier. Make work clearer. Make operations visible.

One continuous workflow — for students, staff and residence administration.

The fragmented reality.

Residence maintenance involves multiple moving parts. A student experiences a problem, the problem must be communicated, the location identified, the right technician reached, the priority set, the work recorded, and the student finally informed.



FIVE OPERATIONAL CHALLENGES

01

Requests become difficult to track.

A student reports an issue without a clear reference or way to follow its progress.

02

Location information is often incomplete.

Personnel need to know exactly where the issue exists. Corridor Hills already has a hierarchy: Block → Floor → Unit → Room.

03

Requests do not always reach the right person.

An electrical fault should not be routed to a plumber. A blocked pipe should not land on a carpenter.

04

Workload becomes difficult to visualise.

Who is available, who is busy, which jobs are waiting, and which are approaching their service deadline.

05

Resolution does not necessarily mean the issue is actually resolved.

A technician may mark a request complete, but the resident may still experience the problem. The platform therefore introduces a student confirmation and reopening mechanism.

THE RESULT

When these conditions persist, maintenance becomes a series of disconnected conversations rather than an operational system.

Built as six blocks, arranged physically, modelled digitally.

STRUCTURE

6

RESIDENCE BLOCKS

Blocks A through D house female students. Blocks E and F house male students. Each unit repeats the same composition — three rooms, two bathrooms and one kitchen.

SIX BLOCKS



A SINGLE UNIT



UNIT POPULATION

3

rooms per unit

2

students per room

6

students per unit

UNIT TYPOLOGY

3 rooms · 2 bathrooms · 1 kitchen · 6 students

FROM PHYSICAL

TO DIGITAL

This structure becomes the foundation of every digital interaction.

Instead of receiving *"the sink in my room is blocked"*, the system can receive:

F·3·01C

F

BLOCK
Block F

3

FLOOR
Third Floor

01

UNIT
Unit 01

C

ROOM
Room C

WHAT THIS ALLOWS

A student residing in **F301C** can be digitally associated with Block F, Floor 3, Unit 01 and Room C. The same code becomes part of every request they raise and every technician assignment that results.

ONE REQUEST

F301C →
Kitchen →
Plumbing →
Blocked sink

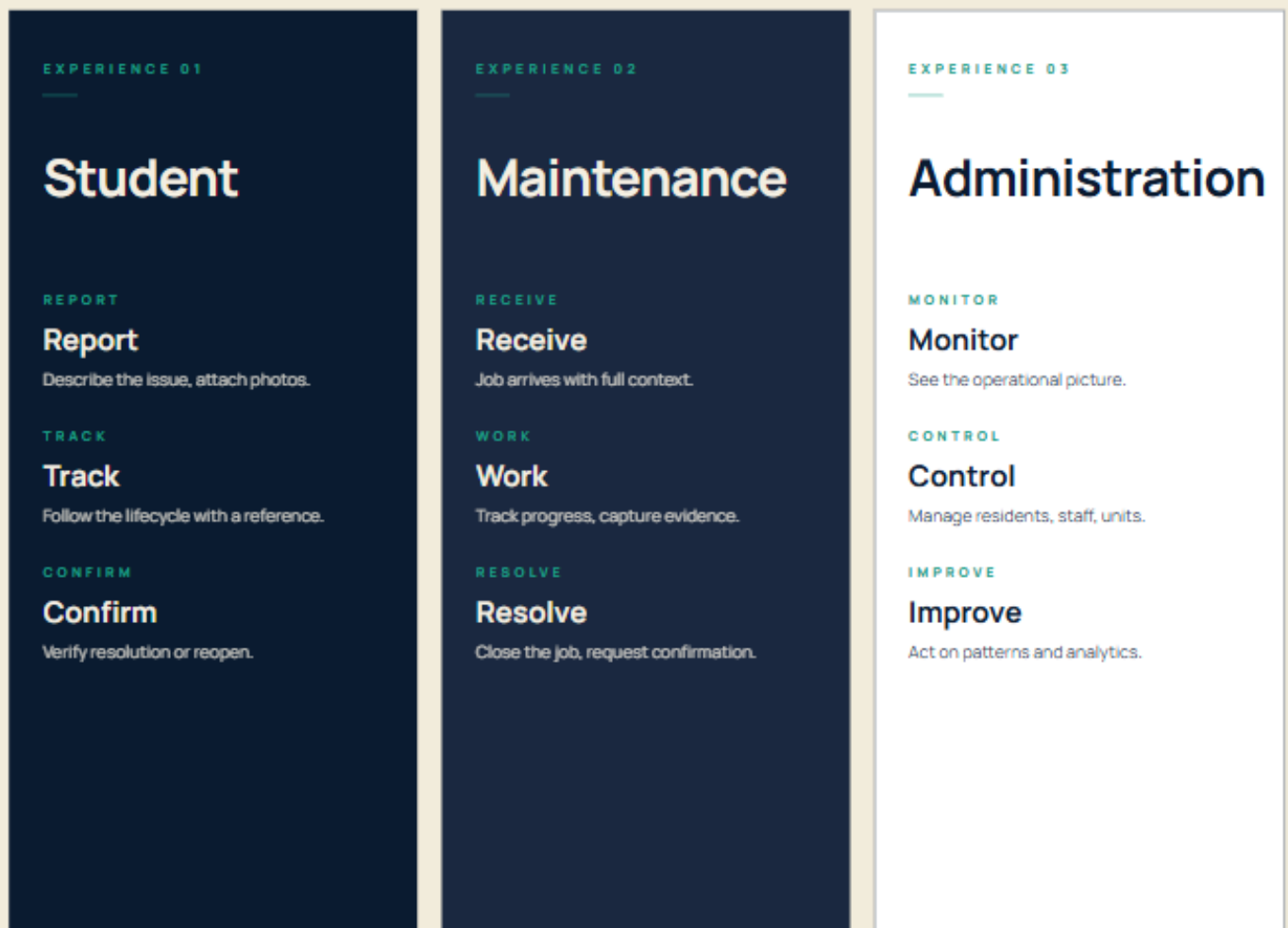
Structured information that drives automation.

CORRIDOR HILLS · BLOCK F, FLOOR 3

A residence begins to exist as data, but never forgets it is a place to live.

From reporting a problem to managing the entire maintenance lifecycle.

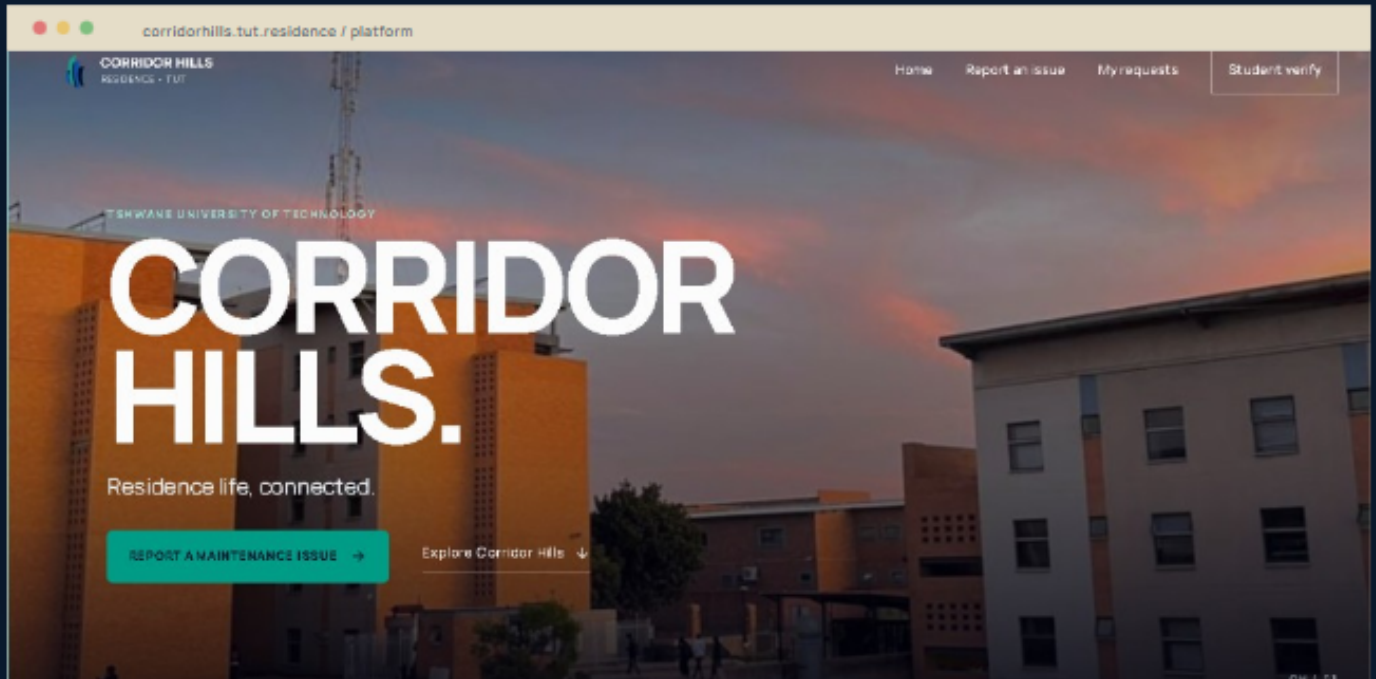
The platform is designed as a Progressive Web Application with three coordinated experiences, each tuned to the role it serves.



ONE PLATFORM

Three experiences, one residence, one continuous workflow.

A landing that already feels like Corridor Hills.



EDITORIAL NOTE 01

Authentic residence identity.

The platform uses Corridor Hills photography and brand identity so students immediately recognise that this is their residence environment.

EDITORIAL NOTE 02

Primary action, immediately visible.

Reporting remains the dominant CTA named REPORT A MAINTENANCE ISSUE, supported by a secondary action to explore Corridor Hills.

NAV

Home · Report an Issue · My Requests · Student Verify

HERO HEADLINE

"Corridor Hills." — Residence life, connected.

SUPPORT LINKS

Track a Request · Explore Corridor Hills · Help

Built for belonging, not just for reporting.

The platform is positioned beyond a maintenance form, towards a residence students recognise as theirs: their photographs, their community, their home.

ng.

is residence life,



CAPTURED INSIDE THE PLATFORM

The community page uses residence own student-life photography — paired with light interactive controls.

EDITORIAL NOTE

“Study. Connect. Show up
for one another.”

— This is residence life, lived together.

PRIMARY

Reporting remains the dominant action

SECONDARY

Track requests, see residence updates

TERTIARY

Belonging, identity, residence story

The platform tells a story about **where students live**, while quietly enabling a smarter way to maintain it.

Designed to feel deliberate.

The interface privileges restraint over decoration, hierarchy over noise, and a single typeface family used with intention.

TYPEFACE

Manrope

A modern geometric sans — clear from 13px metadata to 88px headlines.

FIVE PRINCIPLES

01

Restraint

One accent. One typeface. One voice.

02

Hierarchy

Headlines lead, body supports, metadata frames.

03

Mobile-first

Students reach the platform most often from a phone.

04

Simplicity

The primary action must always remain visible.

05

Motion

Communicates progress, not decoration.

A SINGLE LINE OF INTENTION

The interface should feel less like a form, and more like a product that is genuinely useful.

PRIMARY TYPEFACE

Manrope

PRIMARY ACCENT

Teal #1BA089

tone

Calm · Confident · Operational

The residence itself is the access control.

The system uses Corridor Hills' existing residence records as the source of truth. A student's official student number is mapped to the unit they inhabit. No anonymous reporting, no open forms.

VERIFICATION FLOW

STEP 01

Unit

Enter the unit the student occupies.

STEP 02

Room

Identify the room within that unit.

STEP 03

Student Number

Enter the official student number.

STEP 04

Database Validation

The system matches academic record.

RESULT

Verified.

Access to reporting, tracking and confirmation flows.

ALTERNATE

Access Denied

Records do not align — the system responds quietly, never exposing another resident's data.

A WORKING EXAMPLE

STUDENT NUMBER	220123456
UNIT	F301
ROOM	C
RESIDENCE	F301C

PRIVACY POSTURE

Verification rejects quietly.

The system never reveals whether a different student number exists for the unit, never suggests valid alternative inputs, and never displays another resident's data.

"We couldn't verify those details. Please check your unit, room and student number."

Eight steps. Designed for clarity.

The student experience is deliberately simple. At every moment the resident knows where they are, what is being asked, and what comes next.

01

VERIFY

Confirm residence.

Unit, room and student number.

02

LOCATION

Identify the area.

Room, bath, kitchen or common.

03

TYPE

Choose a category.

Electrical, plumbing or other.

04

DESCRIBE

Describe in words.

Short, simple language works best.

05

EVIDENCE

Capture photos.

In-app camera, multiple images.

06

REVIEW

Confirm summary.

A complete review before submitting.

07

SUBMIT

Submit.

A unique reference appears.

08

TRACK

Follow lifecycle.

Reference follows the student.

REFERENCE NUMBER STRUCTURE

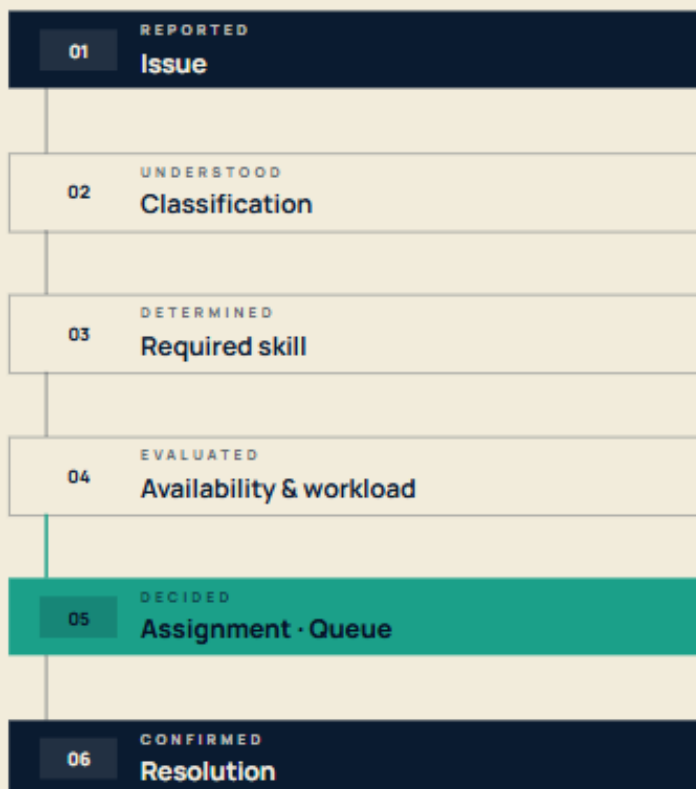
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Each request receives a permanent, traceable identifier used throughout its lifecycle.

Behind a simple form, an automated engine.

The student experience is intentionally simple. What runs underneath is structured, considered, and aimed at routing the right technician to the right place with the right context.

ASSIGNMENT FLOW



THE ENGINE CONSIDERS

REQUIRED SKILL

Electrical, plumbing or general — the request matches a skill set.

AVAILABILITY

Active, idle, on leave — only eligible staff are selected.

CURRENT WORKLOAD

Open headcount maintains balance across the team.

PRIORITY

High, medium, low — informs queue position.

QUEUE AGE

Longest-waiting requests rise to stay fair.

SLA RISK

Requests approaching service targets draw attention.

Every step is auditable.
Every decision is traceable.

Three skill domains, each routed with intention.

Initial maintenance capabilities are organised around three skill domains. Each request finds the technician whose craft matches what is broken.

DOMAIN 01

Electrical

Power, lighting and socket-related faults handled by the residence electrician.

EXAMPLES

Light bulbs
Switches · Sockets
Stoves
Electrical faults

DOMAIN 02

Plumbing

Water-flow and drainage work handled by the residence plumber and the maintenance roster.

EXAMPLES

Sinks · Toilets
Showers · Taps
Blocked pipes
Leaks

DOMAIN 03

General

All physical repairs that do not require a specialist, addressed by general maintenance staff.

EXAMPLES

Beds · Desks · Chairs
Blinds · Curtains
Carpets · Flooring
Doors · Locks

FUTURE EXPANSION

Administrators may configure additional skill domains. The MVP intentionally remains tight; future enhancements may include HVAC, security, painting and beyond.



Unavailable technicians should not pause the system.

Requests wait their turn rather than vanish. When capacity opens, the engine evaluates the queue and assigns the next eligible request automatically.

SCENARIO — STOVE FAULT

REPORTED

“My stove stopped working this morning.”

THE ENGINE DECIDES

Category identified:
Electrical.

THEN IT ASKS

Which technicians carry this skill?
Who is currently available?
What is each technician already doing?

YES

Technician becomes the assignee automatically.

NO

Request is placed in the Electrical queue. It waits.

When a technician becomes available, the queue is re-evaluated and the next eligible request is assigned.

THE ELECTRICAL QUEUE

Requests do not vanish
when capacity is full.

FAULT

Stove · E201-3

NEXT →

FAULT

Light · E102-2

02

FAULT

Socket · B104-1

03

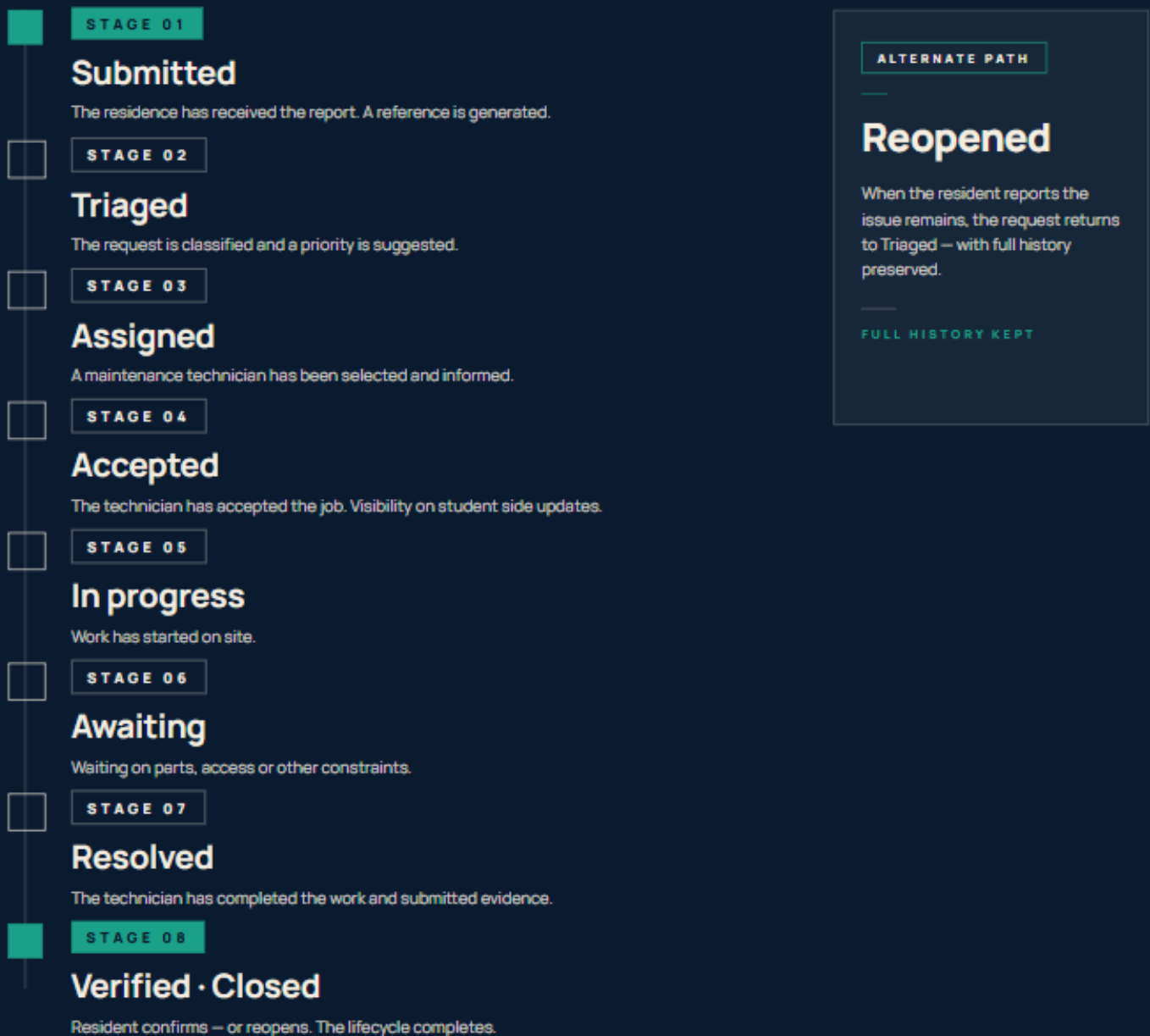
ON RELEASE OF CAPACITY

The engine reassigns
automatically.

No manual chase. No lost work.

Every request travels a traceable path.

A maintenance request is rarely a single moment. It is a sequence with states, transitions and clear ownership at each step.



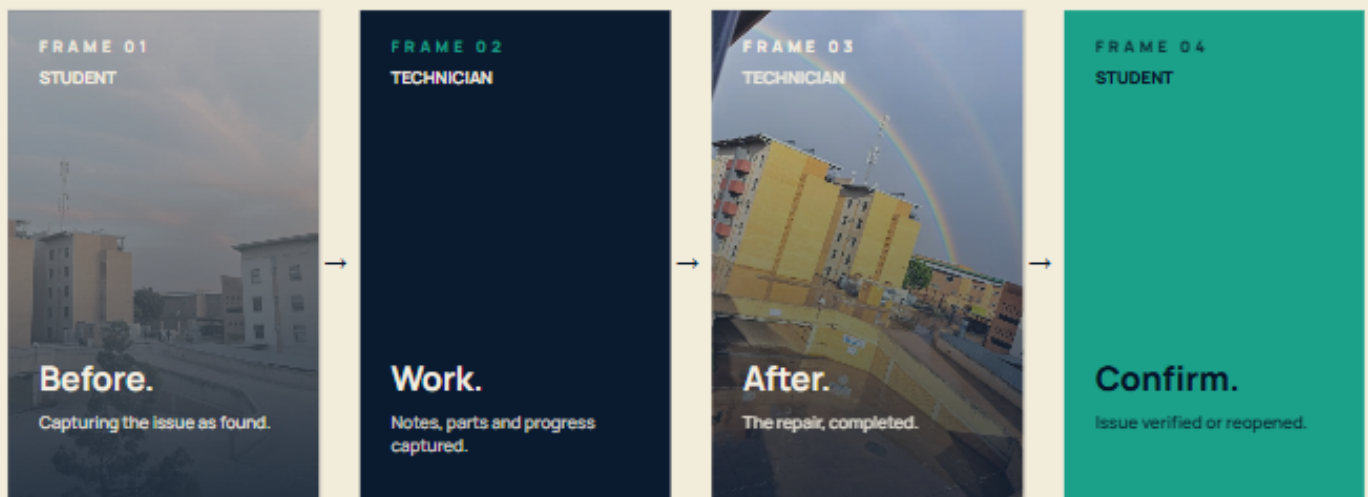
DESIGN NOTE

Transitions are explicit. No state is silent. Every change is timestamped, attributable, and recoverable in audit history.



Camera, work, camera, confirm.

Photographic evidence accompanies the request, the work and the resolution, creating a maintenance record that is harder to dispute and easier to learn from.



WHY THIS MATTERS

Photos reduce ambiguity,
speed up resolution,
and create a record
the residence can learn from.

DESIGN POSTURE

Images are stored securely, attached to the request, and become part of the residence's maintenance history. The system prefers evidence over interpretation.

Before → Work → After → Confirmed.
A record no longer depends on memory.

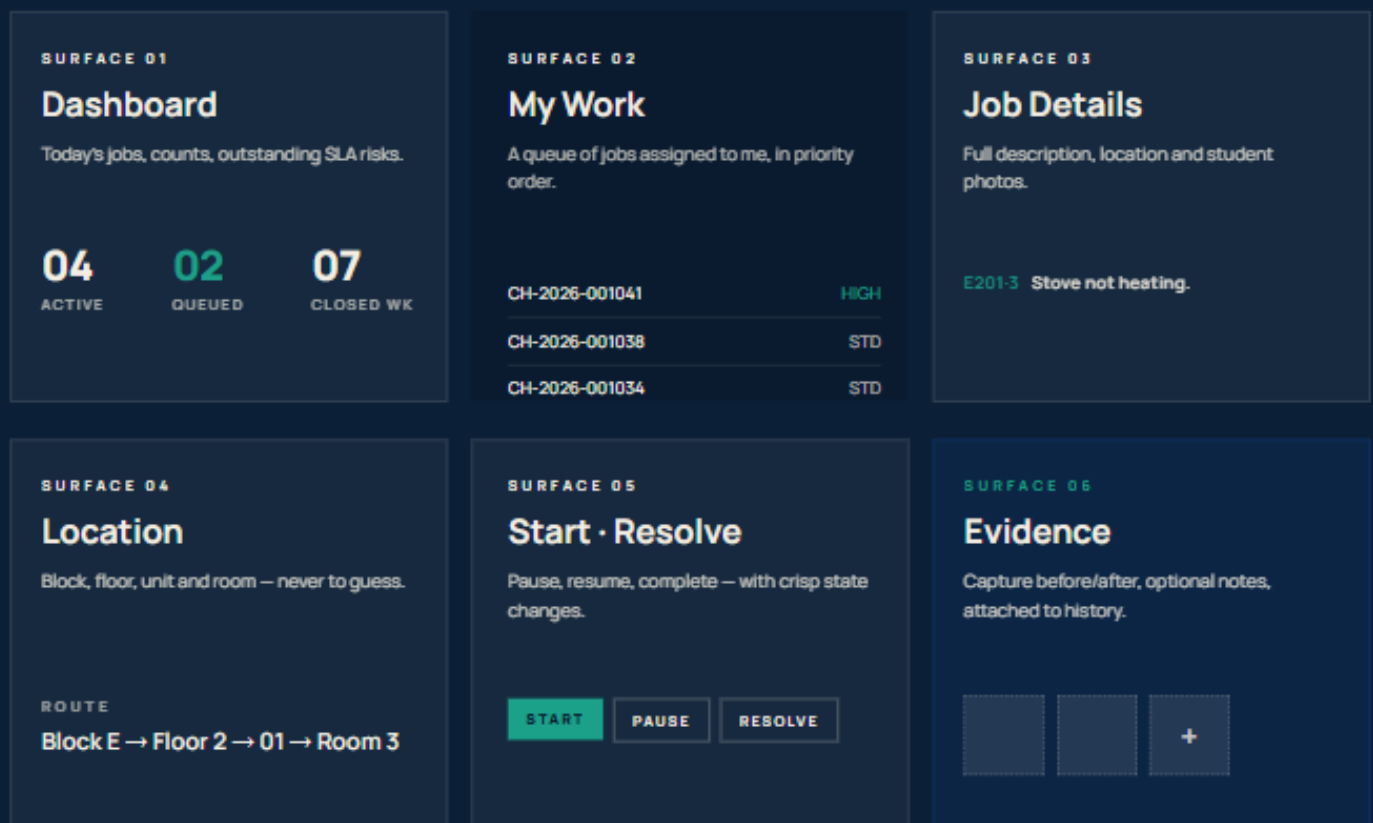
Operational, not ornamental.

The maintenance interface is a working tool. Every element earns its place by helping a technician act faster with less guesswork.

LABEL

Proposed Staff Experience

Conceptual screens shown for stakeholder review. Implementation pending.



NOTE

The staff interface is intentionally quieter than the student landing — fewer decorative elements, faster interactions, and less abstraction.



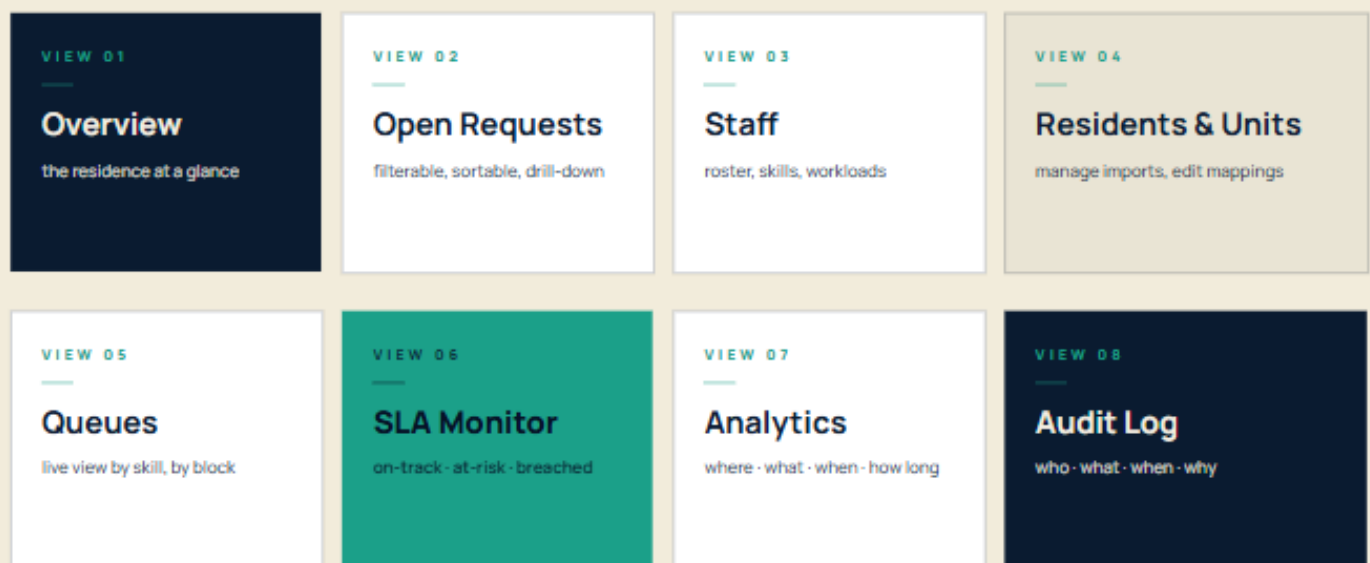
A control surface that sees the entire residence.

The administrator operates differently. They need the picture, trends, blocks at risk, recurring issues, and an audit trail that defends every decision.

LABEL

Proposed Admin Experience

Conceptual screens for stakeholder review.



POSTURE

The admin view is a working instrument.

It favours clarity over decoration and answers questions, **rather than displaying screens.**

AT A GLANCE

SURFACES

08 proposed views

DEPTH

drill-down + filters

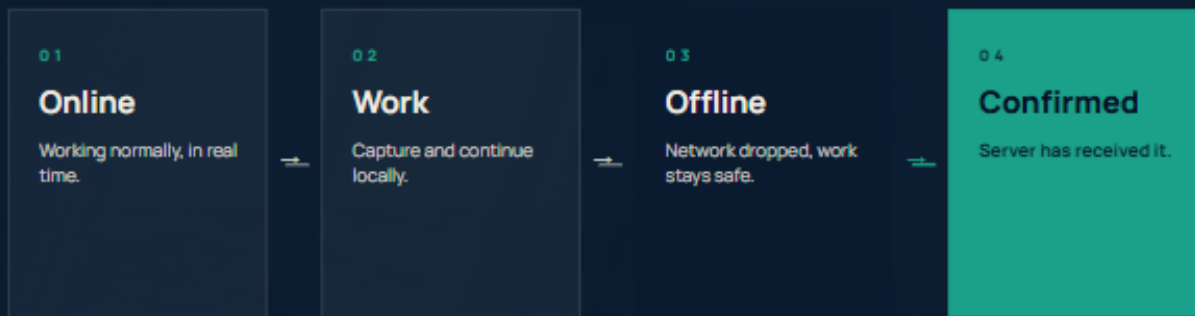
PRINTABLE

export & share

Connectivity drops. Work continues.

Residence environments can be unpredictable — between corners, on certain floors, when the network is congested. The platform is designed to remain useful even when the connection does not.

THE CONNECTION LIFECYCLE



OPERATING POSTURE

The system must never claim an action succeeded until the server confirms it.

DESIGN IMPLICATIONS

Pending actions are queued locally. Conflicts are flagged. The user always knows whether their submission reached the server, and history preserves an honest record of when things happened.

INSTALL-FREE

No app store required.

CROSS-DEVICE

Phone · Tablet · Laptop

RESILIENT

Honest about status

Security is not a feature. It is the foundation.

Security posture is shaped at the architecture stage, not bolted on afterwards. The platform balances access, audit and minimisation across every layer.

PRINCIPLE 01

Resident Verification

Only authorised students from authoritative records may report issues.

PRINCIPLE 02

Role-Based Access

Student, Staff and Admin remain clearly separated at every boundary.

PRINCIPLE 03

Server-Side Authorisation

No protected operation trusts the client alone. Every request is verified on the server.

PRINCIPLE 04

Data Minimisation

Each user sees only what their role requires and nothing further.

PRINCIPLE 05

Secure File Handling

Uploaded evidence is validated, sized and stored safely.

PRINCIPLE 06

Audit Logging

Assignment changes, status changes, edits — all recorded.

PRINCIPLE 07

Historical Integrity.

Records are not destructively deleted because they are no longer active. History remains reconstructable, defensible, and reviewable by residence leadership.

DESIGN NOTE

These principles shape future iterations as much as the MVP. The architecture is built so future tighter policies do not require rewrites.

POSTURE

**Trust earned, never assumed.
Audit, never abandoned.**

Maintenance data, read as questions.

Once requests exist digitally, they become information. Leadership can ask questions of the residence rather than searching for anecdotes.

CHARTS SHOWN

Conceptual — final figures established during pilot.

QUESTION

Where?

Which blocks experience more issues?



A · B · C · D · E · F

QUESTION

What?

Which issue categories dominate?



PLUMBING · ELEC · GEN

QUESTION

When?

When do requests increase?



QUESTION

How long?

Resolution time, by category.



ELEC · PLMB · GEN

QUESTION

Who?

Which skills are under pressure?



PLUMBING
highest pressure

QUESTION

Why?

Which locations experience recurring issues?

F301 — patterns detected.

FURTHER IMPLICATIONS

RECURRING

Recurring Issues

Surfaces repeated fault patterns by unit — prevention begins.

PREVENTATIVE

Preventative Maintenance

Triggers proactive inspections before failures spread.

SLA

SLA Monitoring

Targets tracked, risks highlighted, breaches reported.

Convert maintenance history into a tool that informs
better operational decisions.

CONCEPTUAL
DATA ESTABLISHED
DURING PILOT

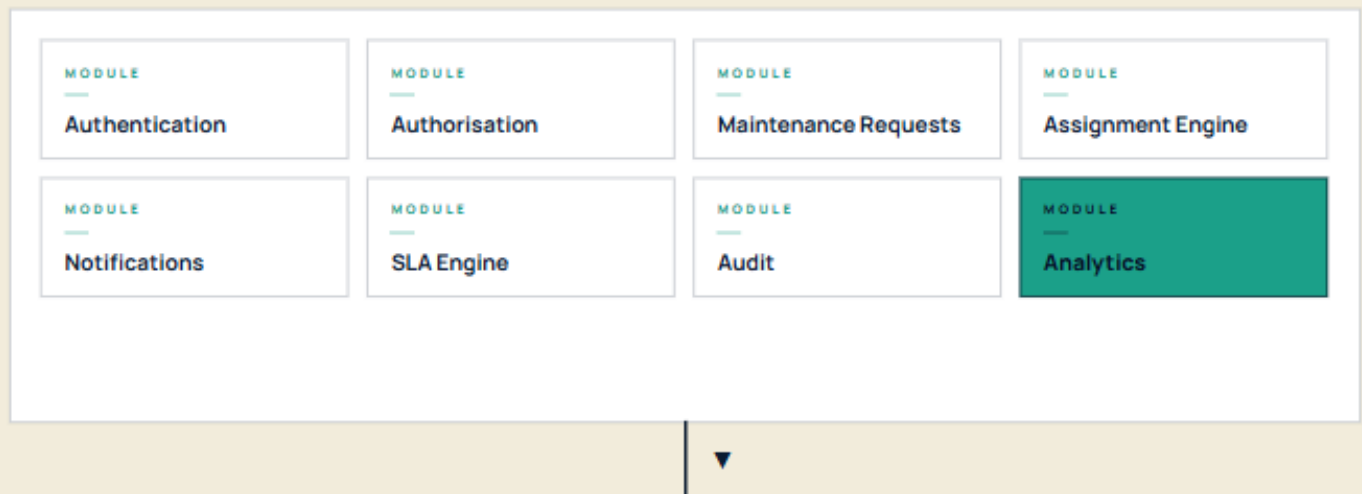
Three layers. One coherent system.

The architecture separates presentation, business logic and data. Each can evolve without dismantling the rest.

LAYER 01 - PRESENTATION



LAYER 02 - APPLICATION / API



LAYER 03 - DATA



DESIGN INTENTION

The architecture is built so future modules can attach without dismantling what already works.
Replacing a layer should not require rebuilding the others.

A disciplined path from idea to operation.

The project follows a Software Development Life Cycle informed by the residence's operational realities. Six overlapping phases — each with a clear purpose, output and review step.

PHASE 01

Requirements

Identify stakeholder, operational, residence and security needs.

OUTPUT · SPEC

PHASE 02

System Design

Define architecture, data models, APIs and UI/UX direction.

OUTPUT · BLUEPRINT

PHASE 03

Development

Build the three experiences, the API layer, and the data model.

OUTPUT · CODE

PHASE 04

Testing

Functional, security, usability, mobile, offline and permission testing.

OUTPUT · VALIDATION

PHASE 05

Deployment

Demonstration environment and production-ready infrastructure.

OUTPUT · LIVE

PHASE 06

Evaluation & Iteration

Feedback from students, staff and admin. Then improve what matters.

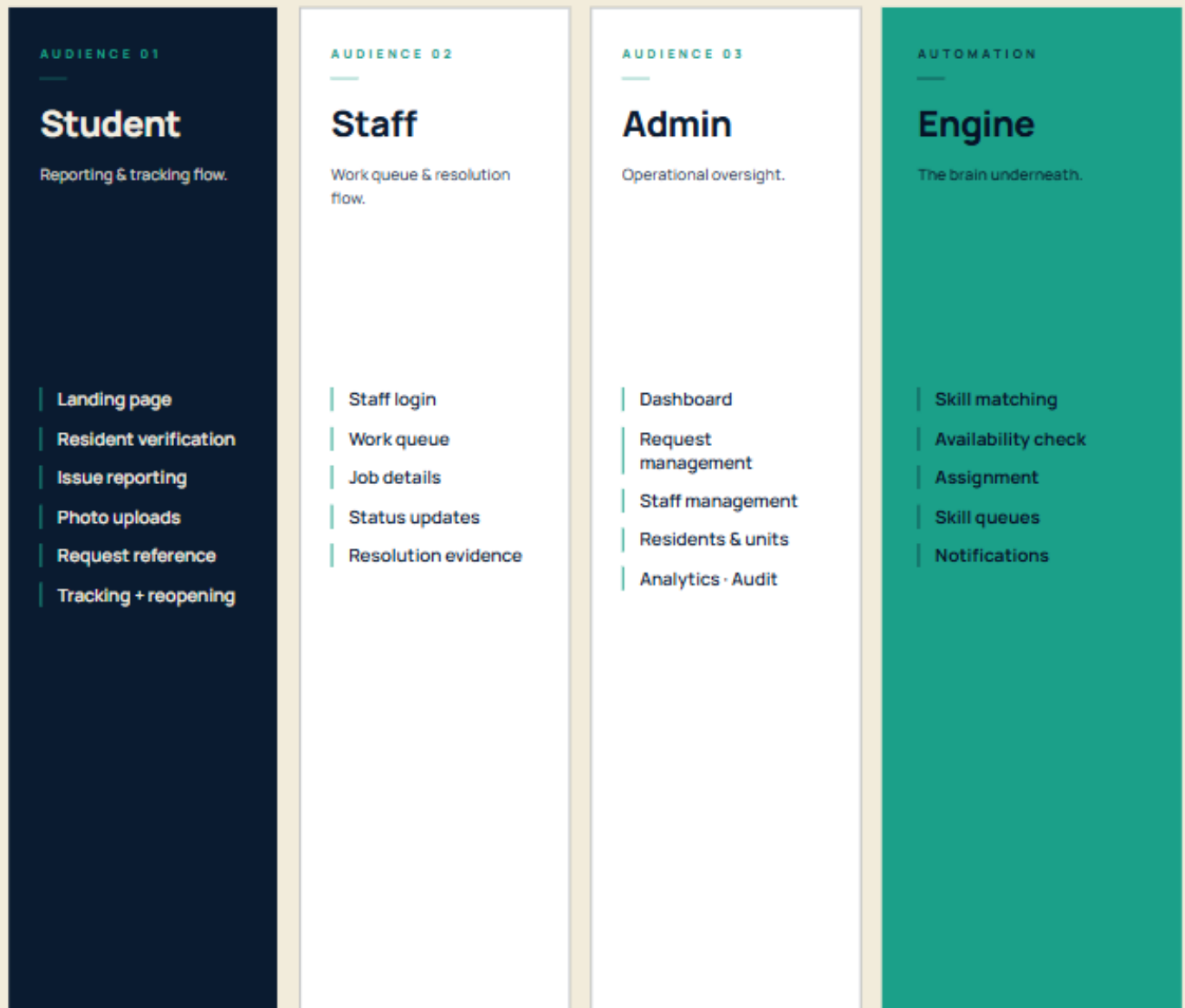
OUTPUT · LEARNING

A NOTE

A demo is not the end — it is the
beginning of operational learning.

Each phase informs the next. Findings during evaluation shape backlog, priorities and the architecture itself — not merely the roadmap.

What ships in the first version — and what explicitly does not.



SCOPE POSTURE

A demonstrable, residence-shaped first version.
Audit-honest. Engineered to grow.

Intentionally out of scope — **intentionally planned.**

The architecture is built so the residence can grow into these capabilities without rewrites. They are not promises — they are informed possibilities.

STATUS

Future · Planned

Distinct from MVP scope.

CAPABILITY 01

**Preventative
maintenance
scheduling**

CAPABILITY 02

**QR / NFC location
identification**

CAPABILITY 03

**Richer maintenance
analytics**

CAPABILITY 04

**Predictive maintenance
signals**

CAPABILITY 05

**Residence asset
management**

CAPABILITY 06

**Maintenance inventory
& parts**

CAPABILITY 07

**Institutional systems
integration**

CAPABILITY 08

Multilingual support

CAPABILITY 09

**Advanced reporting &
exports**

POSTURE

**The MVP will not pretend
these features exist.**

They are listed to communicate the residence's possible future — they will be implemented only after the proven core is in residents' hands and behaves as intended.

Indicators available to evaluate the platform.

The project can be evaluated against operational outcomes that matter to students, maintenance staff and administrators. Baseline and target values will be established during pilot evaluation.

INDICATOR 01

Reporting time

Time required to submit a maintenance request.

BASELINE • TARGET

INDICATOR 02

Visibility

Percentage of active requests with a trackable status.

BASELINE • TARGET

INDICATOR 03

Assignment time

Between submission and a technician being assigned.

BASELINE • TARGET

INDICATOR 04

Resolution time

Average maintenance resolution time, by category.

BASELINE • TARGET

INDICATOR 05

SLA compliance

Requests resolved within configured service target.

BASELINE • TARGET

INDICATOR 06

User satisfaction

Resident and staff feedback gathered during the pilot phase.

QUALITATIVE

AN HONEST NOTE

No performance numbers are presented as achieved.

Specific values, baseline comparisons and quantified outcomes will be established **through pilot evaluation** alongside Corridor Hills residence administration.

**Measurement should follow proof —
not precede it.**



Make maintenance
easier to report.

Make work
easier to manage.

Make residence operations
easier to understand.

PROJECT IDENTITY

Corridor Hills Residence

Digital Maintenance & Residence Operations Platform

REPORT · TRACK · RESOLVE

corridor-hills-residence.vercel.app

Demonstration environment, hosted and accessible.

github.com/Drey780822/corridor-hills Google Drive – docs & materials

Repository containing working artefacts.

Reference documents & supporting material.